

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

Bhaichara
ON
TOP.

STRUCTURE OF ATOM

LECTURE- 8 CHEMISTRY

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Topics To Covered

- 1 CONCEPT OF ORBITALS & SHAPES OF ORBITALS**
- 2 RULES OF FILLING ORBITALS**

REVISION.

① Principal Q. No.

→ n (shell).

$n = 1, 2, 3, 4, \dots$

Energy = $-13.6 \frac{Z^2}{n^2}$ eV/atom.

$E_n = a \cdot \frac{n^2}{Z}$

→ $\bar{e} = 2n^2$

2. AZIMUTHAL Q. No.

→ l (subshell).

For n , $l = 0$ to $(n-1)$

$n=1$, $l=0$ (s).

$n=2$, $l=0, 1$ (s, p).

$n=3$, $l=0, 1, 2$ (s, p, d).

$n=4$, $l=0, 1, 2, 3$ (s, p, d, f).

⋮

Magnetic Q. No.

→ m_l (orbital).

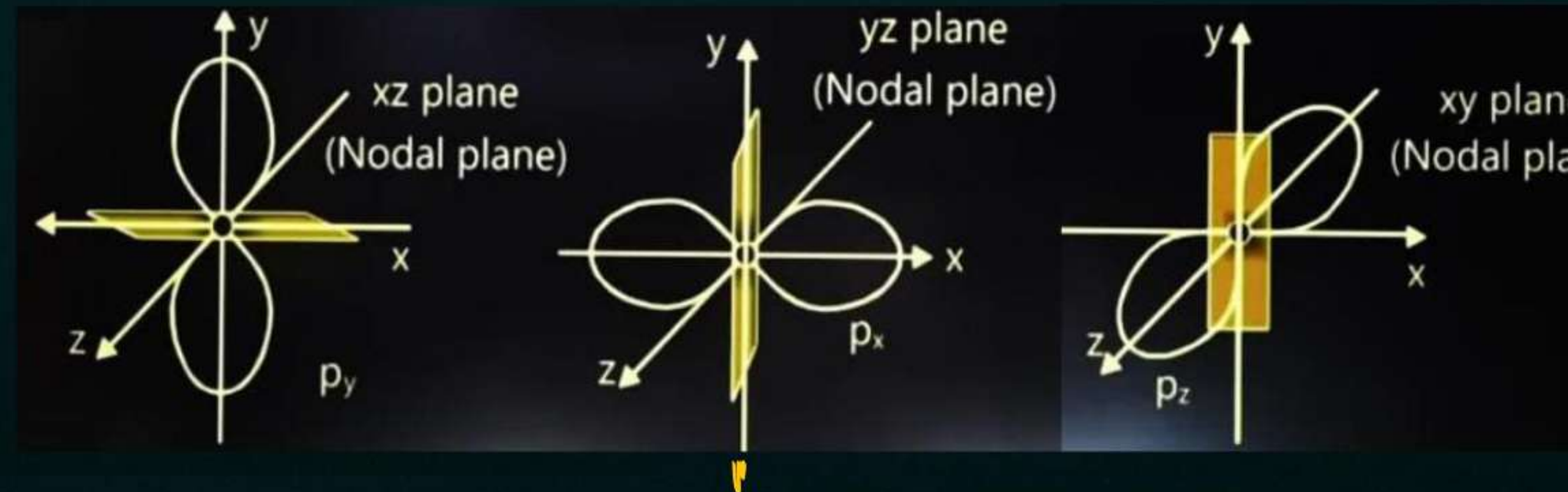
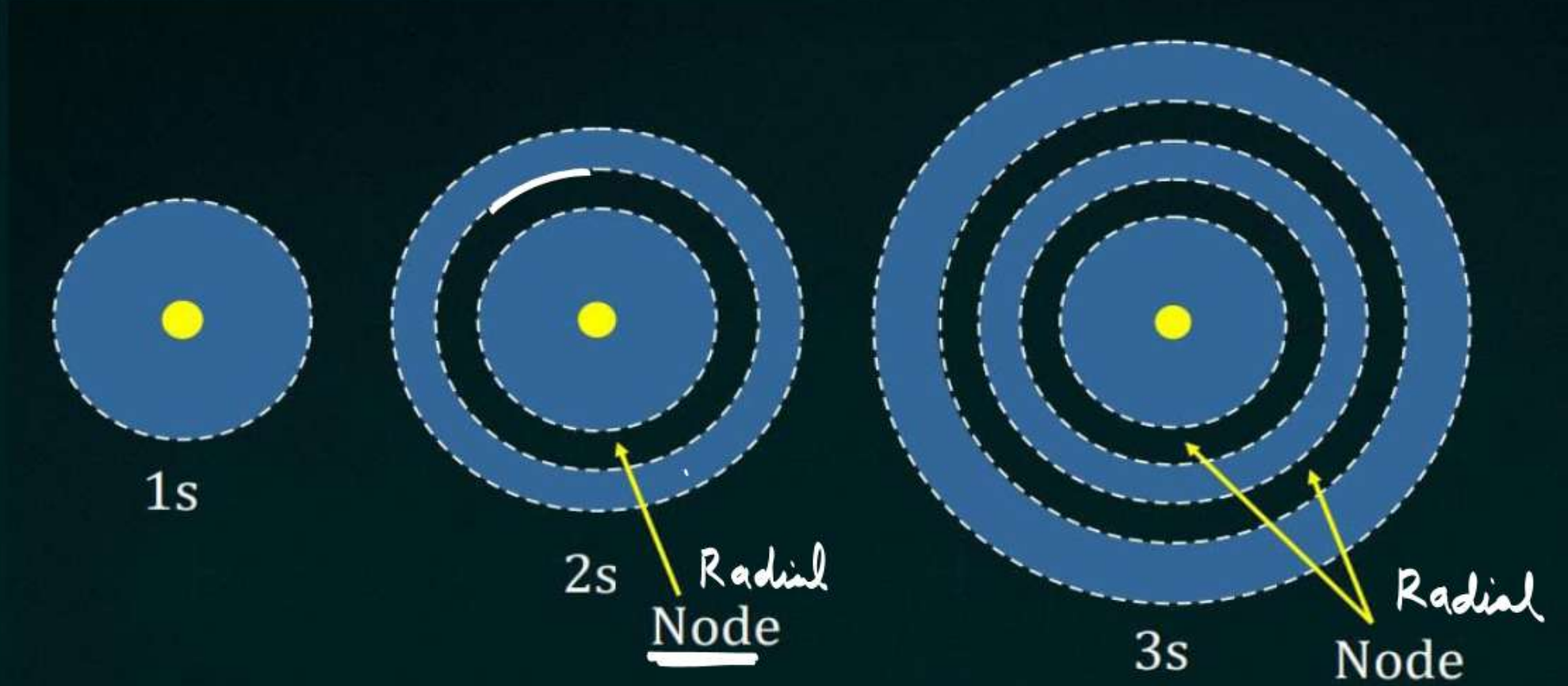
For l , $m_l = -l$ to $+l$

$l=0$, $m_l=0$ → one orbital.

$l=1$ (p), $m_l = -1, 0, +1$ → 3-orbital.

$l=2$, (d), $m_l = 5$ -orbital

$l=3$, (f), $m_l = 7$ -orbital.



4.) SPIN QUANTUM NUMBER.

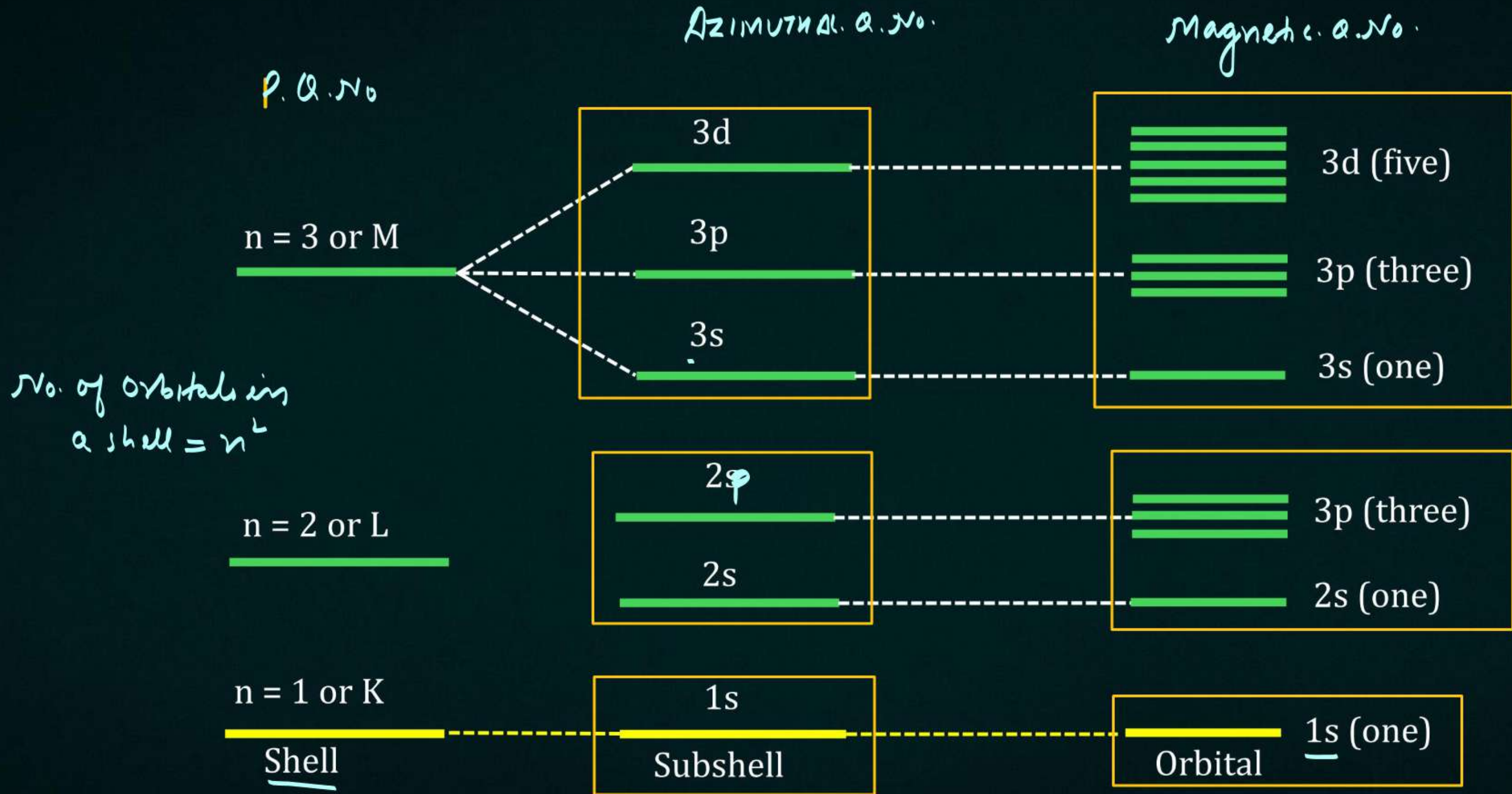
It provides us the information about the spin of e^- in orbital. It is denoted by 's' or m_s .

Spin quantum no. has only two values.
i.e. $+\frac{1}{2}$ & $-\frac{1}{2}$.

For clockwise spin, $s = +\frac{1}{2}$ ($\uparrow$)
& for anticlockwise spin, $s = -\frac{1}{2}$ ($\downarrow$)

NOTE $\rightarrow$

QUANTUM NUMBERS
Bahut Important
ha "JIGAR".

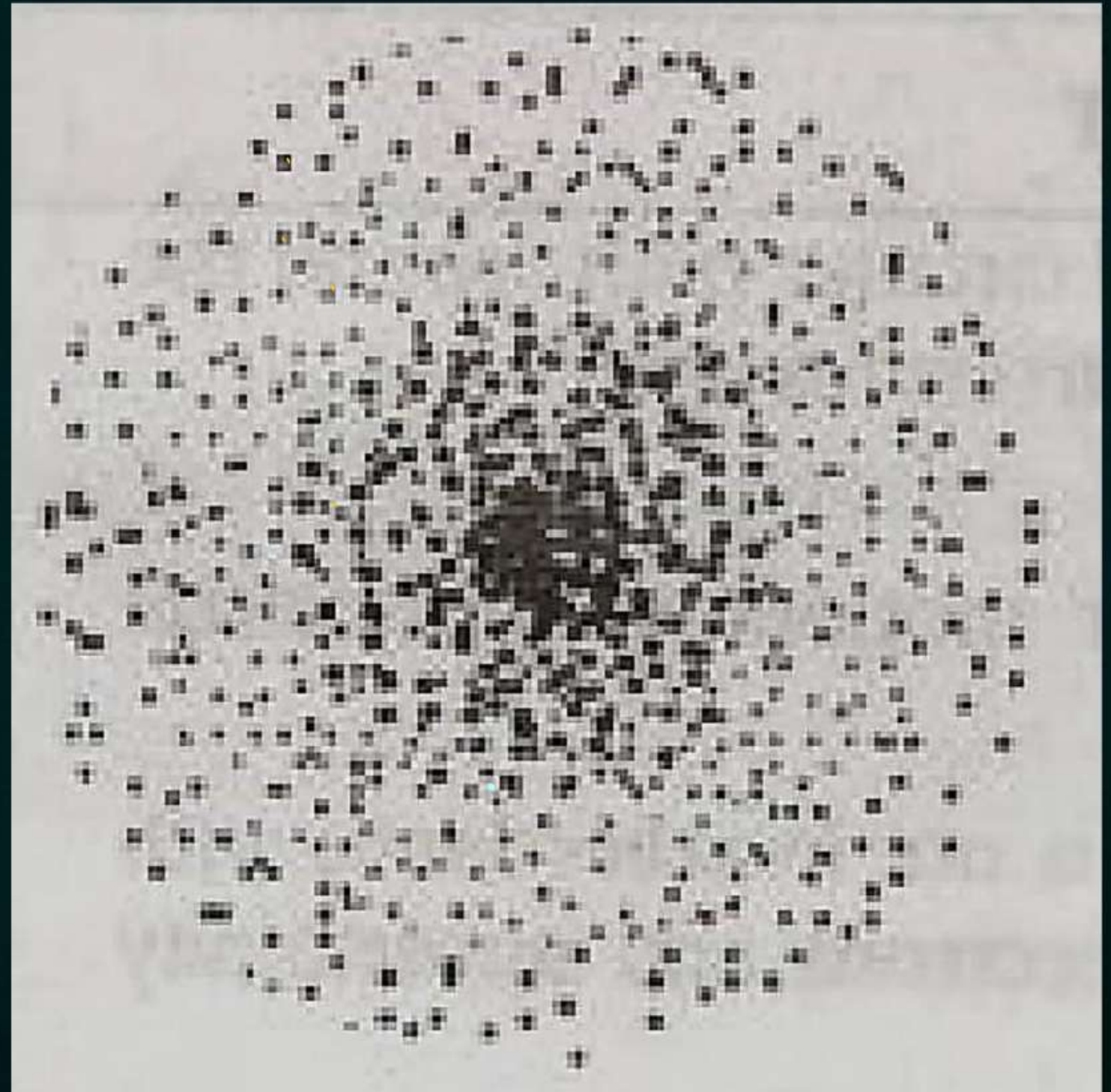


CONCEPT OF ORBITAL



Concept of orbit was discarded by Heisenberg's uncertainty principle. i.e. we can't describe the exact position of e^- but we can talk about the probability of finding an e^- around the nucleus.

Orbital is defined as "3D space around the nucleus where probability of finding an e^- is maximum ($\approx 90\%$)".
When ψ^2 probability is expressed in form of dots then it is named as electron cloud.



SHELL (ORBIT). ✓



SUBSHELL



ORBITAL. ✓

SHAPES OF ORBITALS



1) Shape of s -orbital

On the basis of probability calculations from quantum mechanics, probability of finding an e^- in s -orbital is equal in all directions around the nucleus.

i.e. s -orbital has spherical shape.

- We know that for s -subshell, $l=0$, $m_l = -l$ to $+l$
 $= 0$
i.e. one orbital.

- s -orbital is non-directional.

Note

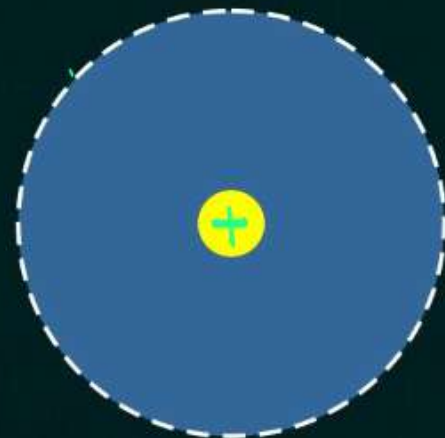
Shape is determined with the help of Quantum mechanics.

SHAPES OF ORBITALS

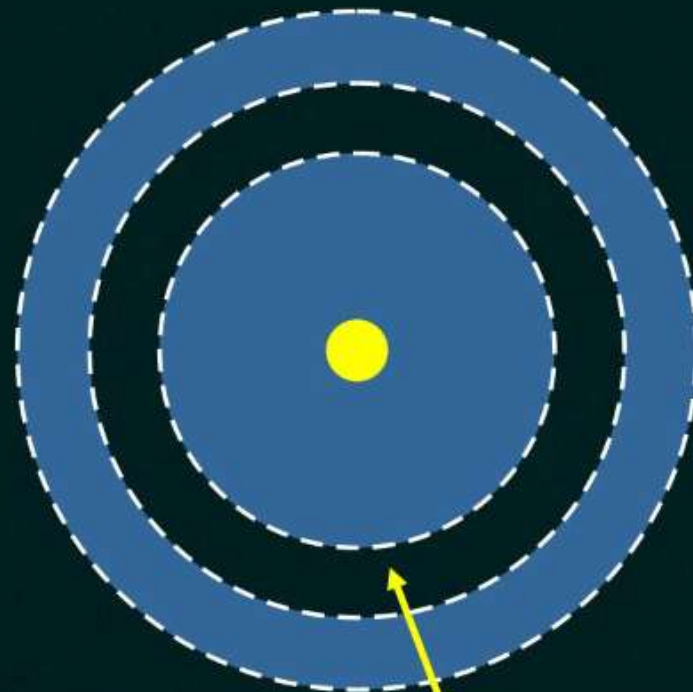


- Shape of $1s, 2s, 3s, 4s$ has same shape
'the only difference is size.'

$1s < 2s < 3s < 4s \dots$

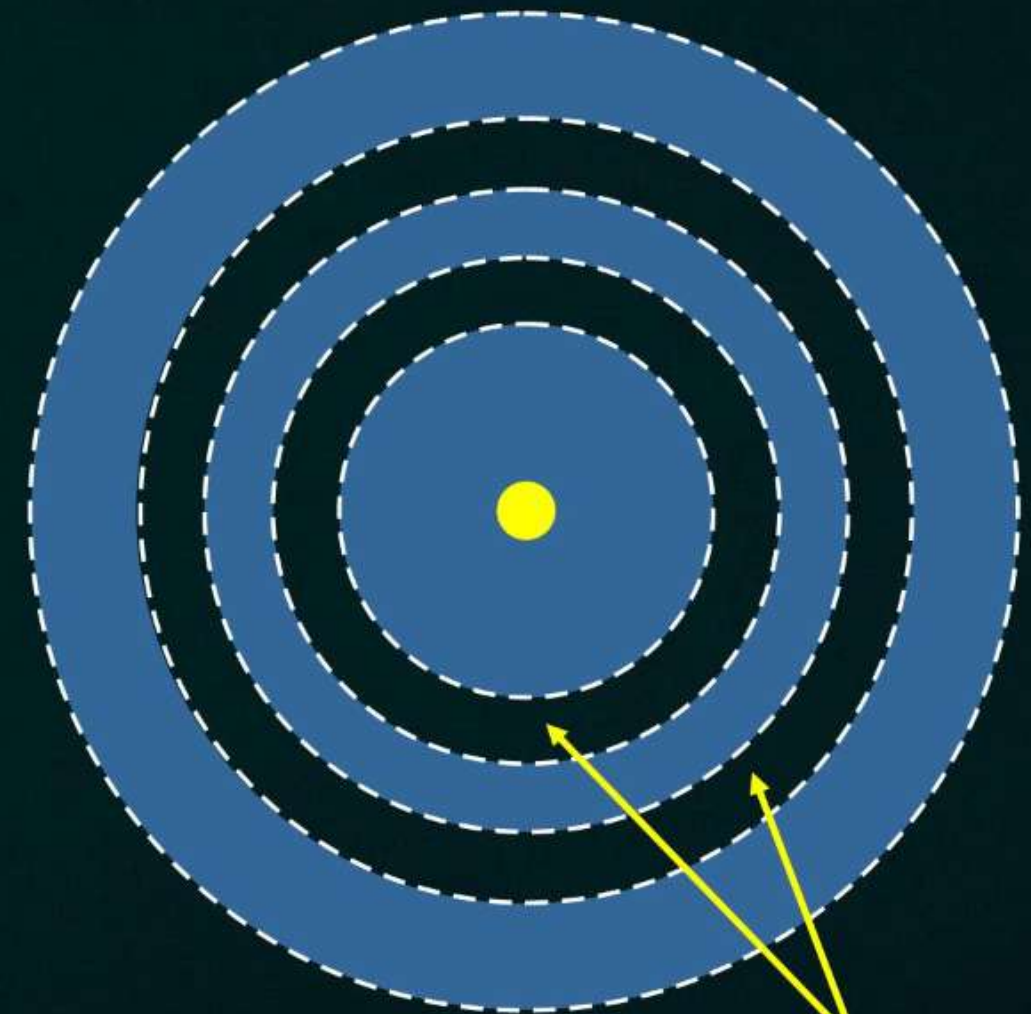


$1s$



$2s$

Node



$3s$

Node

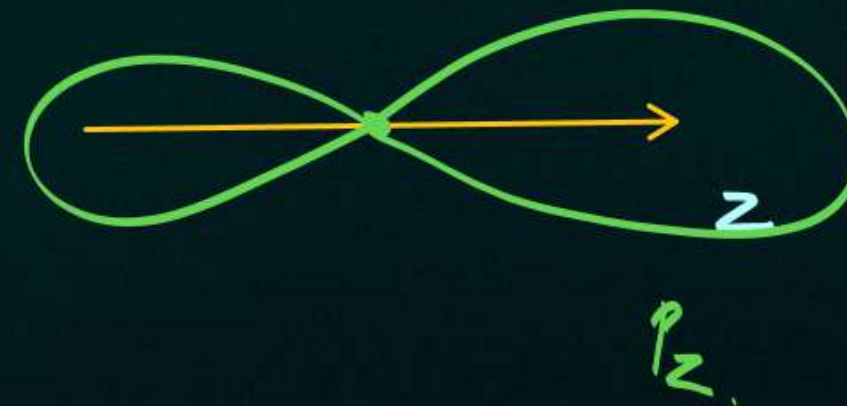
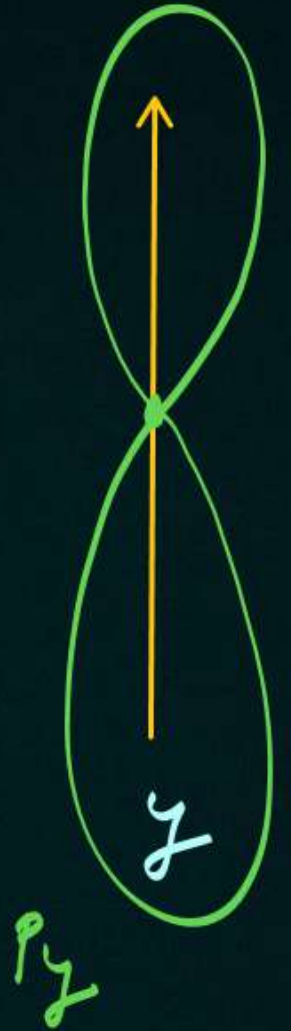
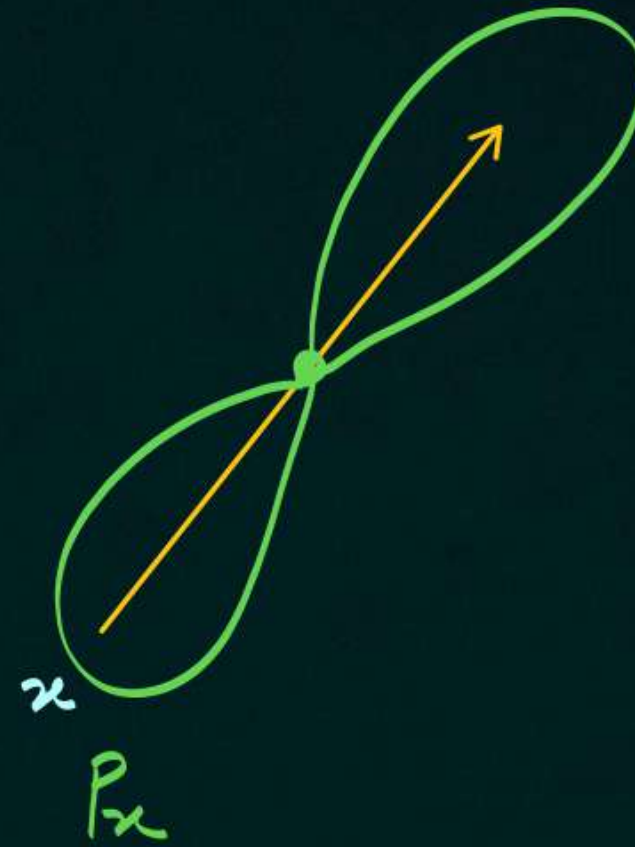
SHAPES OF ORBITALS



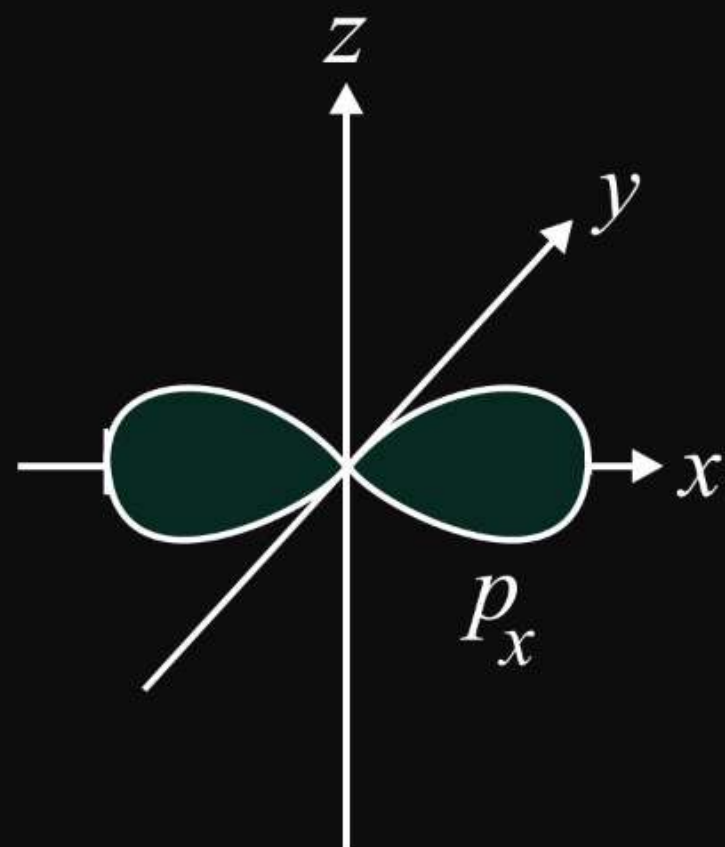
2). Shape of p-orbital : → On the basis of probability

Calculations, probability of finding an e^- in a p-orbital is found to equal in two opposite lobes around the nucleus. i.e. p-orbital has dumbbell shape.

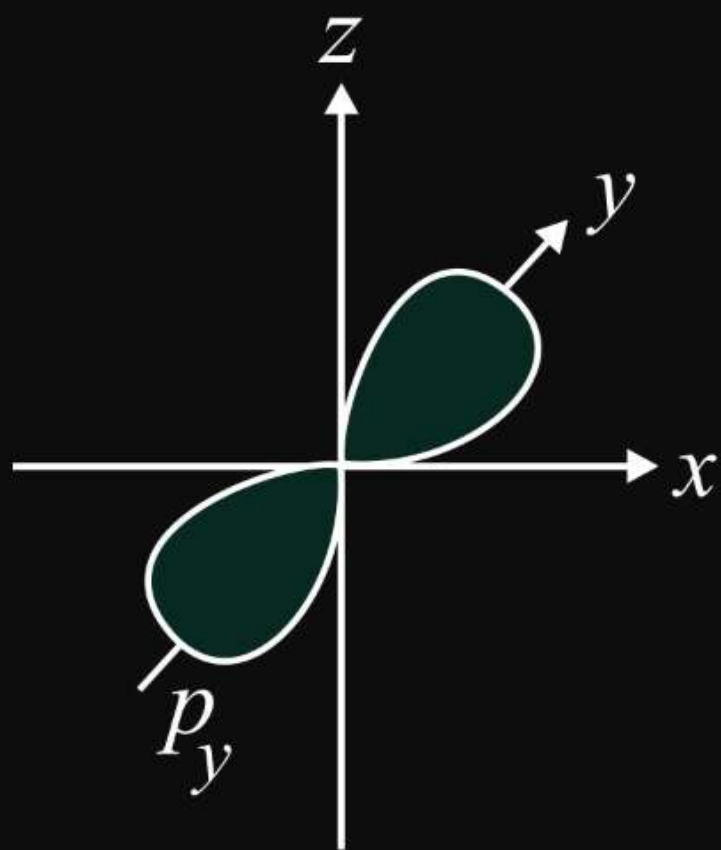
- For - p-subshell, $l=1$, $m_l = -1, 0, +1$ i.e. 3-orbitals, namely, p_x, p_y, p_z .
- P-orbitals are directional in nature.



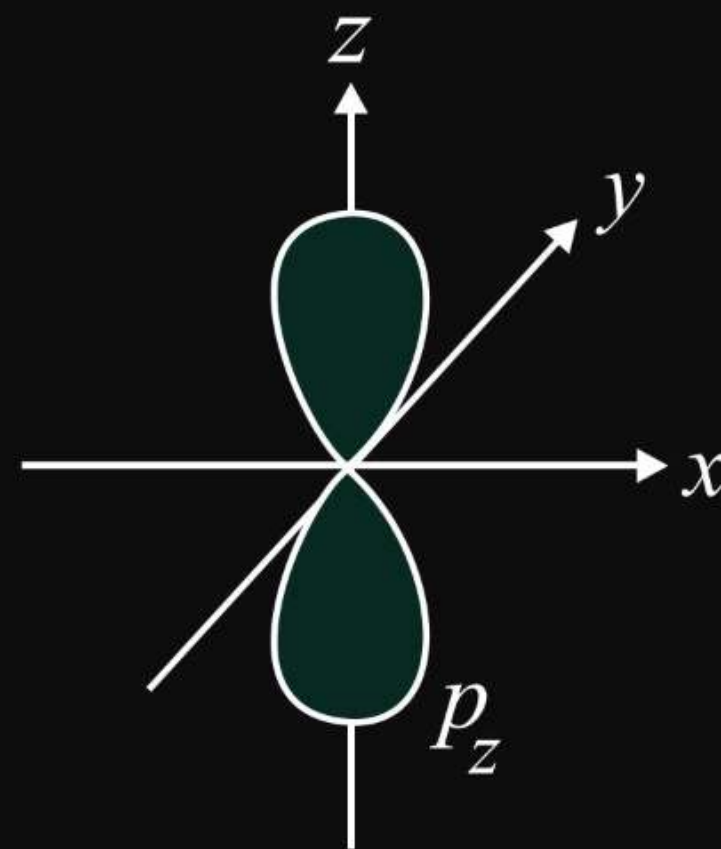
SHAPES OF ORBITALS



p_x orbital



p_y orbital



p_z orbital

SHAPES OF ORBITALS



3) Shape of d-orbital →

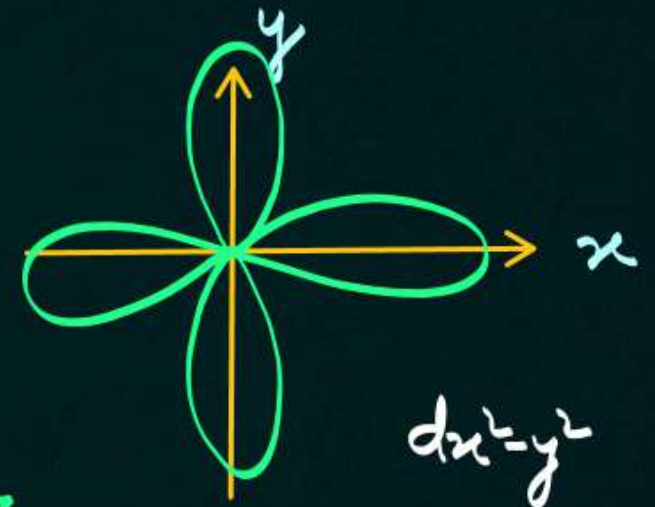
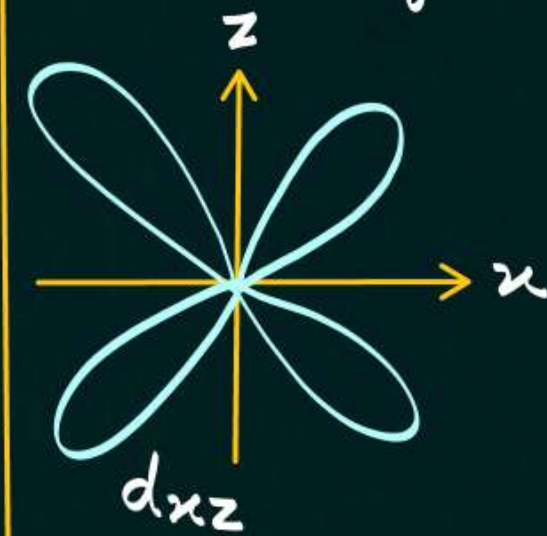
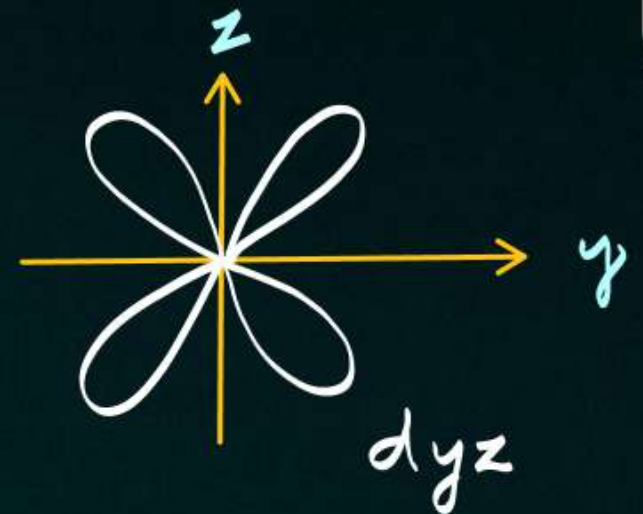
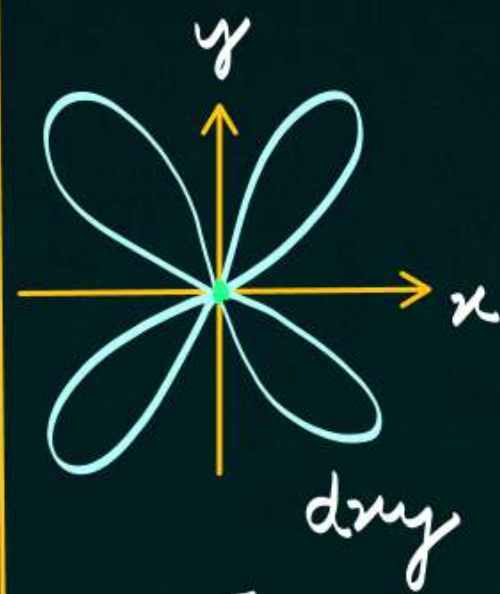
On the basis of probability calculations, the probability of finding an e^- in d-orbital is found to be maximum in four lobes around the nucleus.

i.e. Shape of d-orbital is found to be double dumbbell or clove leaf shape.

For d-subshell, $l = 2$, $m_l = -2, -1, 0, +1, +2$ i.e. 5-orbitals namely

d-orbital is directional.

$d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$
Non-Axial Axial



(dough-nut shape.)

RADIAL AND ANGULAR NODES

- 1) **Radial Node or Spherical Node**: $\rightarrow$ 3D space inside the orbital where probability of finding the e^- is least.
- 2) **Angular node or nodal plane**: $\rightarrow$ Plane passing through the nucleus where probability of finding the e^- is least.

Formula

$$\text{Radial node} = n - l - 1$$

$$\text{Angular node} = l$$

Q. Calculate the radial & angular node for 3d.

$$\text{Sol. Radial node} = \frac{n - l - 1}{3 - 2 - 1} = 0$$

$$\begin{aligned} \text{Angular node} &= l \\ &= 2 \end{aligned}$$

QUESTION



Calculate the total number of angular nodes and radial nodes present in 3p orbital.

COMMUNITY

① Z^2

② Wahed Rasool.

③ Dr. Aleem.

DIFFERENCE BETWEEN ORBIT AND ORBITAL

M.W
Zarur likhna.



ORBIT	ORBITAL
1. An orbit is a well defined circular path around the nucleus in which the electrons revolve.	1. An orbital is the three dimensional space around the nucleus within which the probability of finding an electron is maximum (upto 90%).
2. It represents the planar motion of an electron around the nucleus.	2. It represents the three dimensional motion of an electron around the nucleus.
3. The concept of an orbit is not in accordance with the wave character of electrons and uncertainty principle.	3. The concept of an orbital is in accordance with the wave character of electrons and uncertainty principle.
4. All orbits are circular and disc like.	4. Different shapes. orbitals have different i.e. s-orbitals are spherically symmetrical, p-orbitals are dumbbell shaped and so on.
5. Orbits do not have any directional characteristics.	5. All orbitals except s-orbitals have directional characteristics.
6. The maximum number of electrons in any orbit is given by $2n^2$ where n is the number of the orbit.	6. The maximum number of electrons present in any orbital is two.

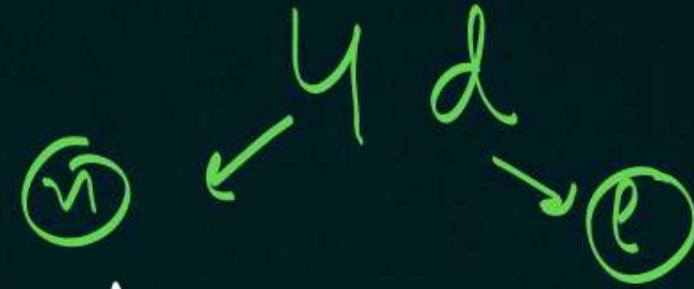
Q. Calculate the radial nodes for 4d orbital?

a) 0

b) 1

c) 2

d) 4



Hint: $\rightarrow$

$$\text{Radial node} = \underline{n} - \underline{l} - \underline{1}$$

$$4 - 2 - 1$$

$$= 4 - 3 = 1$$

Q. Which of the following d-orbital is axial.

a) dx_y

b) dyz

c) dz^2

d) dx_z

Bhalchava on Top.

Q. Which of the following Orbital is non-directional.

a) s

b) p ✓

c) d ✓

d) f ✓

REVISE KAR LENA.

Thank
You